



Clinical Trials DB

Clinical Trials Database

— User Manual —

*A role-based application for managing
clinical trial information.*



Database Systems Project

INTRODUCTION

Clinical Trials Database is a Java-based system created to store, organize and consult information related to clinical trials. The application manages data about trials, patients, doctors, hospitals and patient descriptions in a centralized database.

Its main objective is to make clinical trial information easier to access and manage. The system connects all the main entities through a relational database and provides different options depending on the user role.

Through this role-based structure, Trial Managers, Doctors, Patients and Guests can use the application according to their permissions, making the system more organized, secure and efficient.

SYSTEM OBJECTIVE

Objective	Description
 Centralize data	Store clinical trial information in one structured database.
 Connect entities	Link trials, patients, doctors, hospitals and descriptions through relationships.
 Control access	Show different options depending on the user role.
 Support management	Allow users to consult, update and analyse clinical trial data efficiently.

These objectives make the application a clear, organized and role-based environment for managing clinical trial information securely.

SYSTEM REQUIREMENTS

These are the minimum requirements needed to run the application correctly and connect to the database.

Requirement	Description	Status
Java environment	Required to run the application.	☐ Checked
SQLite database	Stores the clinical trial information.	☐ Checked
Eclipse or compatible IDE	Recommended for development and testing.	☐ Checked
Valid credentials	Needed to access role-based menus.	☐ Checked
Database files available	The ClinicalTrials.db file must be connected correctly.	☐ Checked

FIRST STEPS

When the application starts, the user sees the main access menu. From this menu, the user can log in, sign up, enter as a guest, or exit the system. After access, the system checks the user role and opens the corresponding menu with the available options.

Application access flow

The system identifies the user before showing the available options.

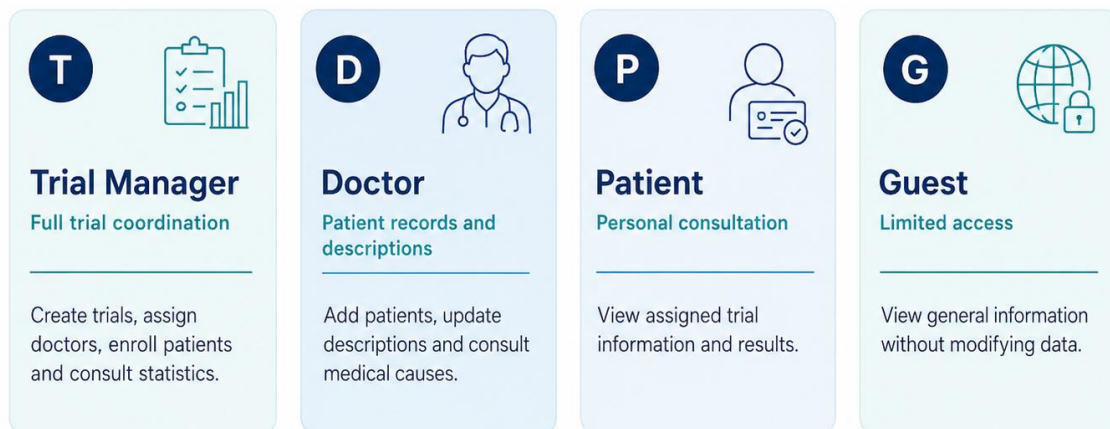


ACCESS TO THE APPLICATION

The application provides different access options from the main menu. Users can log in with an existing account, create a new account, enter as a guest, or exit the system. Each option guides the user to a different access path depending on their needs and permissions.

Access option	Description
Login	Allows registered users to enter their username and password. If the credentials are correct, the system opens the corresponding role menu.
Sign Up	Allows new users to create an account and be assigned to a specific role.
Guest Access	Allows users to enter with limited permissions and consult only general information.
Exit	Closes the application safely.

ROLE-BASED ACCESS



Role-based design

The interface is designed around permissions. This means that users do not all see the same options: each role receives a menu adapted to its responsibilities within the clinical trial workflow.

PERMISSION MATRIX

The permission matrix summarizes the actions available for each user role. This helps users understand which operations they can perform inside the system and which actions are restricted.

Feature	Trial Manager	Doctor	Patient	Guest
Create / remove trials	✓	✗	✗	✗
Assign doctors to trials	✓	✗	✗	✗
Enroll patients	✓	✓	✗	✗
Manage patient descriptions	Limited	✓	✗	✗
View own trial information	✓	✓	✓	Limited
Consult statistics and reports	✓	Limited	✗	✗

This permission structure keeps the application organized and protects clinical information by limiting each user to the actions related to their role.

TRIAL MANAGER MANUAL

The Trial Manager is the main coordinator of the Clinical Trials Database. This role has the highest level of access and is responsible for managing the clinical trial workflow.

Trial Managers can create and remove trials, assign doctors, enroll patients, consult trial details and generate useful reports. This role connects the main entities of the system: trials, doctors, hospitals, patients and descriptions.

Function	Description
Create a new trial	Adds a new clinical trial to the database.
View all trials	Displays all registered clinical trials and their details.
Assign doctors to trials	Connects doctors with specific clinical trials.
Enroll patients in trials	Adds patients to selected trials.
Consult statistics and reports	Compares results, predicts how many new patients are needed, and calculates average duration and average budget.
Remove data	Removes patients from trials or deletes trials when needed.

Function Table

Trial ID • Trial name • Starting date • Duration in days • Budget • Target patients

Information required to create a trial

This role ensures that clinical trials are correctly registered, connected with the necessary medical staff and monitored through reports and statistics.

DOCTOR MANUAL

The Doctor role is focused on patient-related information. Doctors can add patients to existing trials, manage patient descriptions and consult medical information related to patient causes.

This role does not control the full trial configuration, but it supports the clinical workflow by keeping patient data and medical descriptions updated.

Function	Description
Add patient to trial	Adds a new patient to an existing clinical trial.
Add patient description	Creates a medical description including gender, cause and patient ID.
Update patient description	Modifies the information stored in an existing patient description.
Remove patient description	Deletes a patient description when it is no longer needed.
Search patients by cause	Finds patients with the same medical cause.
Consult patient information	Allows doctors to view patient-related data.

Patient ID • Patient name • Result status • Trial ID • Hospital ID • Description ID

**Information
required to add a
patient**

This role ensures that patient information is correctly stored, updated and connected with the corresponding trial, hospital and medical description.

PATIENT MANUAL

The Patient role is designed for consultation purposes only. Patients can view their assigned trial information and check their result status, but they cannot modify or delete any data.

This role allows patients to access only the information related to their participation in the clinical trial process.

Function	Description
View assigned trial	Displays the clinical trial assigned to the patient.
View result status	Allows the patient to consult their current result information.
Consult basic trial information	Shows general information related to the assigned trial.
Consult hospital information	Displays the hospital associated with the patient's participation.

Information available to the patient

Patient ID • Trial ID • Result status • Hospital ID

This role ensures that patients can consult their own clinical trial information in a simple and secure way, without modifying the database.

GUEST MANUAL

The Guest role is designed for users who want to access the application without a registered account. Guests can consult general information about the system, but they cannot modify, insert or delete any data.

This role provides limited access and does not allow users to view private patient information or perform database management operations.

Function	Description
View general information	Allows guests to understand the purpose of the application.
Consult public trial information	Displays limited information about the clinical trial system.
Access without account	Allows basic access without login credentials.
Restricted operations	Prevents guests from inserting, updating or deleting data.

General system information • Limited trial information • No private patient data

Information
available to the
guest

This role ensures that external users can understand the system without accessing sensitive clinical data or modifying the database.

HOW TO USE THE MENU

The Clinical Trials Database uses a console-based menu system. Users select an option by typing the corresponding number and pressing Enter. After each operation, the system shows the result and allows users to continue using the menu or return to the previous section.

Console menu navigation

Initial Menu
1. Login 2. Enter as Guest 3. Sign Up 4. Exit Select an option: _

Trial Manager Menu
1. Create trial 2. View all trials 3. Assign doctor 4. Enroll patient 5. Consult reports 0. Back to main menu Select an option: _

Navigation

rules

Rule	Description
Select an option	Type the number of the desired option and press Enter.
Follow instructions	Enter the information requested by the system.
Use valid IDs	When selecting trials, patients, doctors or hospitals, use the IDs shown by the system.
Confirm operations	Review the information before deleting or modifying data.
Return to menu	After completing an operation, go back to the previous menu or main menu.

DATABASE SECTIONS

The Clinical Trials Database is organized into different sections that represent the main information managed by the system. Each section stores a specific type of data and is connected to the rest of the application through the database relationships.

Section	Purpose
Trials	Stores the main information of each clinical study.
Doctors	Stores doctors and their assigned trials.
Hospitals	Stores hospitals involved in clinical trials.
Patients	Stores patient information and results.
Descriptions	Stores patient medical details such as gender and medical cause.
Users and Roles	Controls login information and access permissions.

These sections help keep the clinical trial information structured, connected and easy to consult.

SECURITY AND PRIVACY INFORMATION

The Clinical Trials Database manages clinical and personal information, so security and privacy are important aspects of the system. Access to the application is controlled through user credentials and role-based permissions.

Each user can only access the options related to their role. This helps protect patient information and prevents unauthorized users from modifying sensitive data.

Security aspect	Description
Login credentials	Users need a valid username and password to access the system.
Role-based permissions	Each user can only access the options assigned to their role.
Patient data protection	Patient information should only be consulted by authorized users.
Restricted guest access	Guests cannot access private patient information or modify the database.
Controlled operations	Insert, update and delete operations are managed through specific Java manager classes.

This security structure helps keep clinical trial information protected, organized and accessible only to the correct users.

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FINAL CONCLUSION

The Clinical Trials Database provides a structured and role-based solution for managing clinical trial information. By connecting trials, doctors, hospitals, patients and descriptions in one database, the system makes clinical data easier to organize, consult and update.

The application also includes controlled access through user roles, useful management functions and a clear console-based interface. Overall, this project combines database design, Java programming and access control to create an organized system for clinical trial management.